$$y'' + \alpha y' + \beta y = 2e^{3x}$$

$$y(x) = 2x^2 e^{3x}$$

$$y'(x) = 2x e^{3x} + 3x^2 e^{3x}$$

$$y''(x) = 2e^{3x} + 6x e^{3x} + 6x e^{3x} + 9x^2 e^{3x}$$

$12x$

$$9x^2 + 12x + 2 + \alpha(3x^2 + 2x) + \beta(x^2) = 2$$

$$\begin{cases} 9 + 3\alpha + \beta = 0 \\ 12 + 2\alpha = 0 \end{cases}$$

$$\int_2^{\infty} \frac{1}{x \log(x)^{\alpha} (e^x + 1)^{\beta}} dx \quad , \quad \int_1^2 \frac{1}{\log(x)^{\alpha}} dx \quad , \quad \log(x) \ll x \quad \int \frac{1}{x^{\alpha}}$$

$$x^{1-\alpha}$$